

A small supervision budget barely improves weak-to-strong generalization

On GPT-2 / BoolQ, mixing a ground-truth budget into the weak labels helps only modestly — and *where and how* it is spent show little effect.

A consistent account: the student largely reproduces the teacher's errors, and ground truth corrects mainly the examples it directly labels — so the effective lever is *volume*, not placement or combination.

GPT-2 family · BoolQ + SciQ · 3 seeds · paired per-(pair,seed) contrasts · pre-registered

Scope & setup

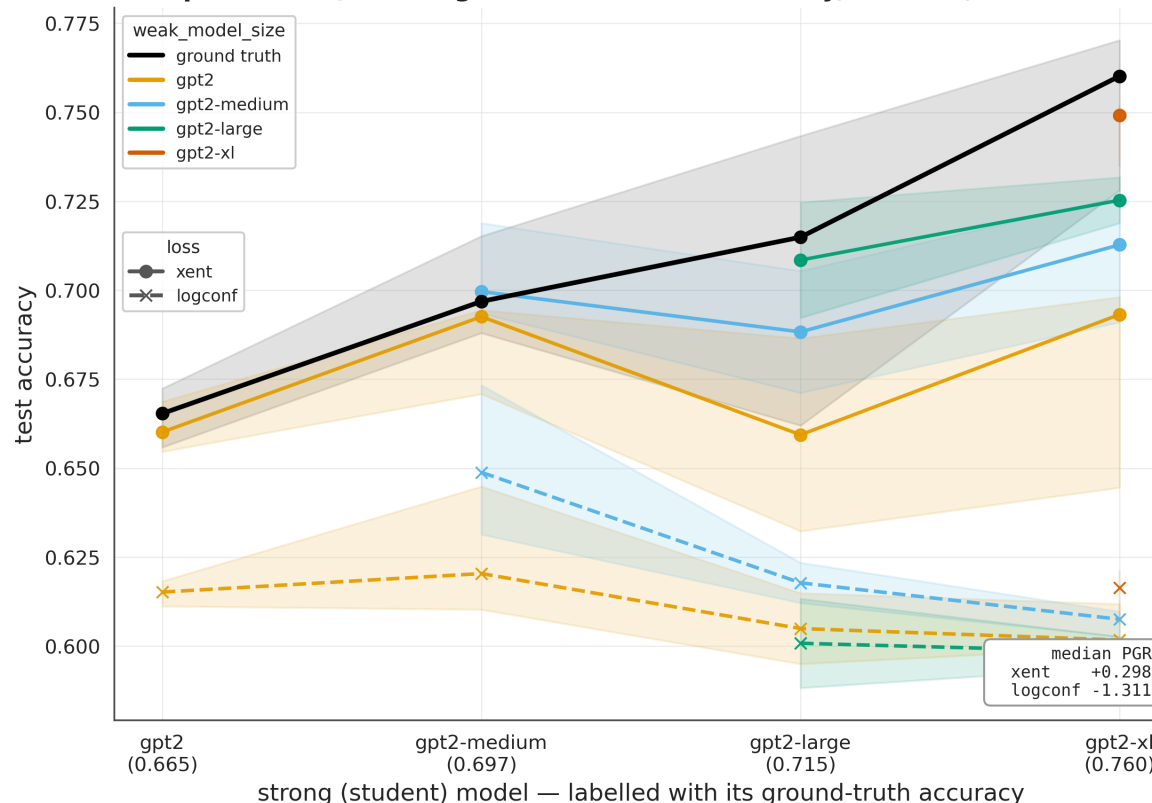
- **Setting:** weak-to-strong generalization — a strong *student* is trained on a weak *teacher's* labels.
- **Extension:** a *supervision budget* — mix a fraction of ground-truth ("strong") labels into the weak labels, and sweep that fraction.
- **Models: GPT-2 family only** — gpt2 / medium / large / xl, with within-family student-teacher pairs.
- **Tasks: BoolQ** (required) + **SciQ** (cross-task check).
- **Readout:** median **PGR** across the model sweep (raw accuracy primary; PGR secondary).

3 seeds · paired per-(pair,seed) contrasts · pre-registered · gpt2-large (seed 1) excluded by rule

GPT-2 sweep at 25% ground truth

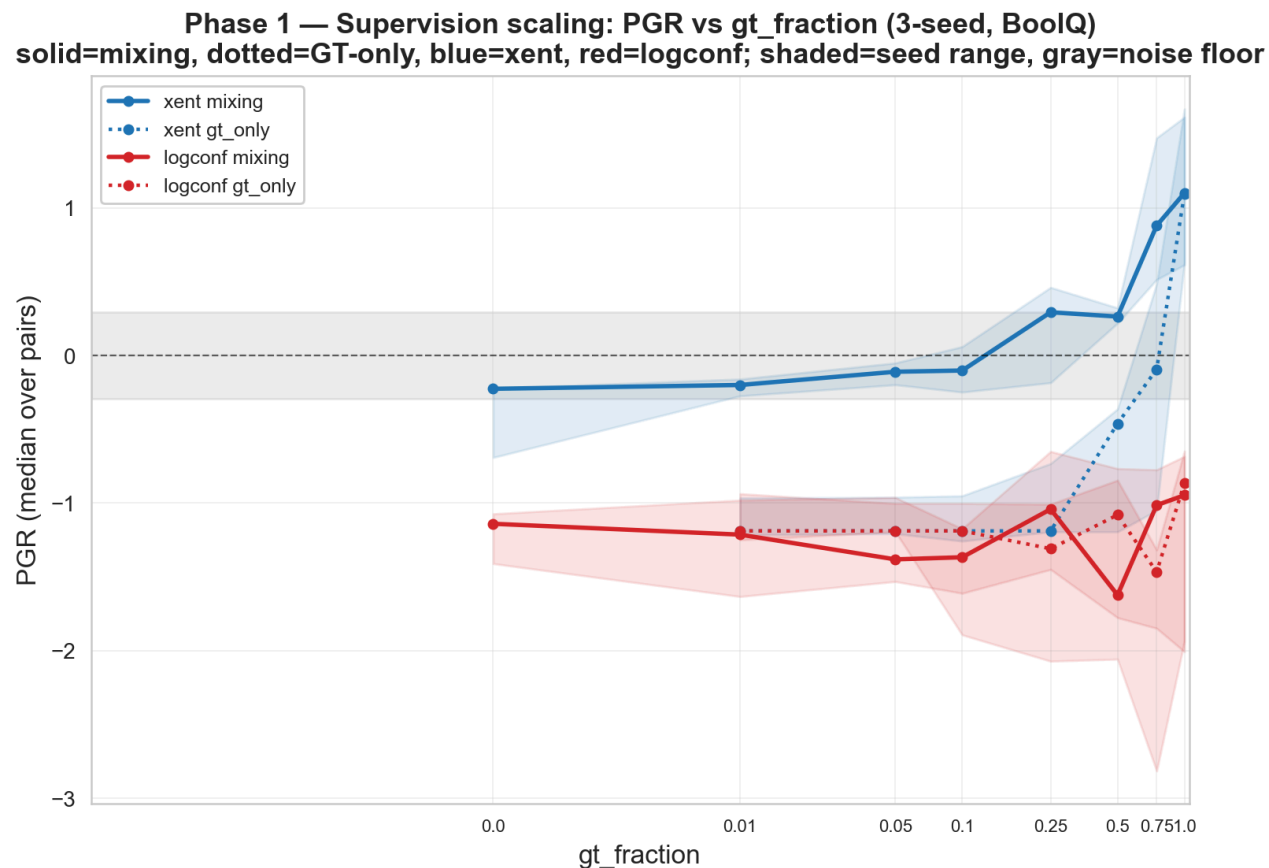
- GPT-2 family, BoolQ, **25% ground truth mixed into the weak labels** — standard sweep format.
- **Median PGR (xent) = +0.30** across the sweep; logconf is deeply negative (-1.3) and is dropped.
- The 0% baseline is **-0.27**; 25% GT moves the median to **+0.30** — a small positive.

W2SG sweep — BOOLQ + 25% ground truth (GPT-2 family, 3 seeds, band = seed range)



How *much*? — back-loaded, and saturating by ~75%

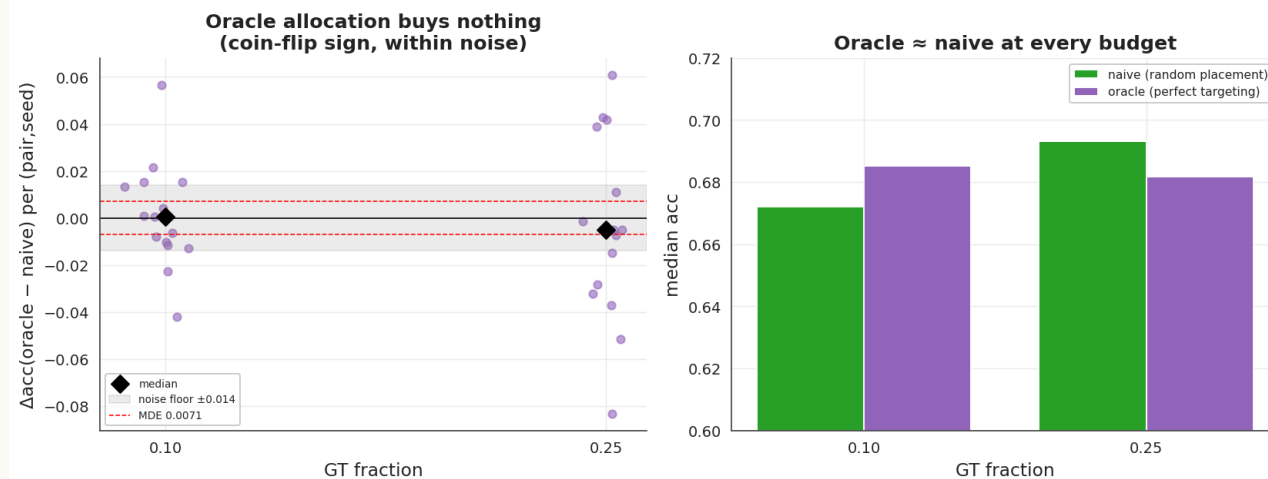
- Up to **10% GT**, the gain over the 0% baseline stays within the **0.014 noise floor**.
- Median xent PGR is **back-loaded**: $-0.22 \rightarrow +0.30$ (0.25) $\rightarrow +0.28$ (0.50) $\rightarrow +0.90$ (0.75) $\rightarrow +1.04$ (1.0).
- The 0.75 point: the **0.50 $\rightarrow$ 0.75 step is the largest**, and **0.75 $\rightarrow$ 1.0 is within noise** — the gain is essentially complete by ~75%.
- A pre-registered concave-knee-at-25% prediction was **refuted**, and retracted after the multi-seed data.



Where it's spent — within noise (allocation null)

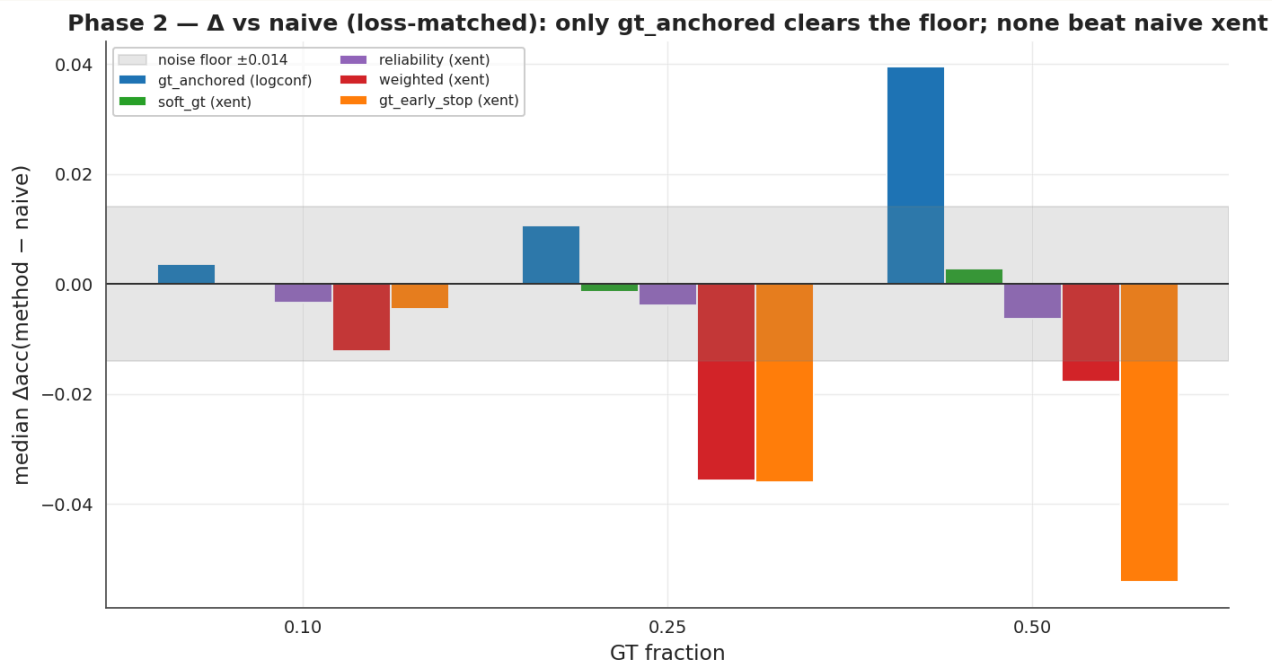
- An **oracle** uses held-out GT to place the budget exactly on the **teacher-wrong rows** — an upper bound on any allocation rule.
- Paired against random placement at matched budget: oracle - random = **+0.0006** (0.10), **-0.0049** (0.25) — both within the MDE.
- If the best possible placement ties random, allocation heuristics have **little to gain** on this testbed.

Phase 1b · B — WHERE you spend GT does not matter (allocation NULL): a perfect error-targeting oracle ties random



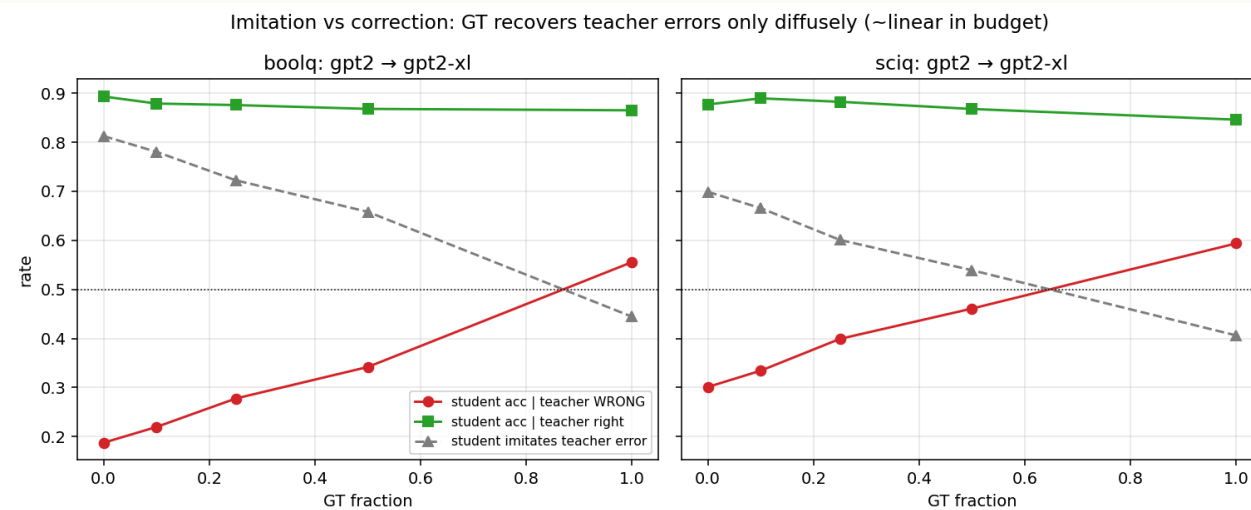
How it's combined — within noise (combination null)

- Five methods vs naive mixing at matched budget: GT up-weighting, soft-GT targets, GT-anchored logconf, reliability-weighted weak labels, GT-based early stopping.
- Median Δ vs naive ≈ 0 across $\{0.10, 0.25, 0.50\}$; none shifts the curve left or raises the ceiling.
- **gt-anchored** is the only method above the floor (+0.040 at 0.50), but it recovers logconf toward xent **without exceeding plain xent** (0.642 vs 0.697).



Mechanism: the student reproduces the teacher's errors

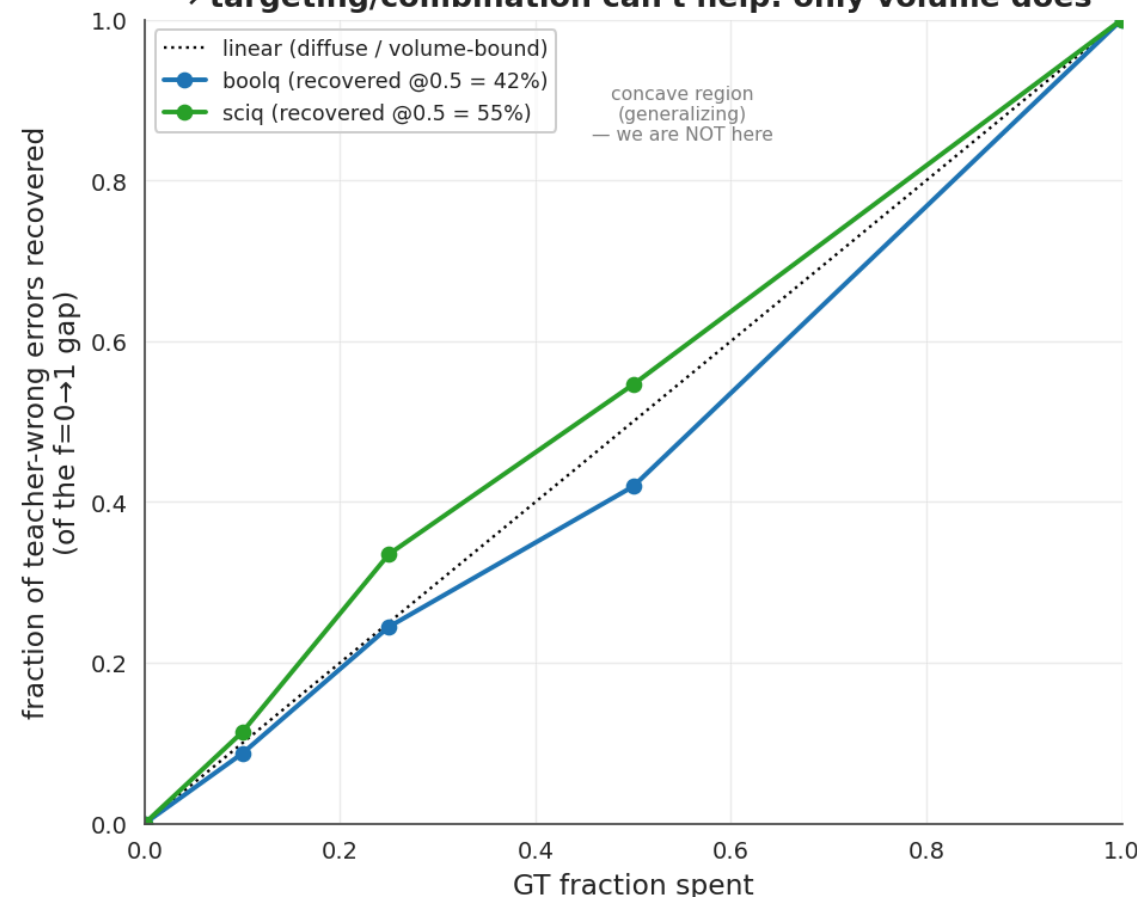
- Probe: gpt2 → gpt2-xl, joining teacher and student **per-example** test predictions.
- On teacher-wrong rows at 0% GT, the student reproduces the teacher's wrong answer **81% (BoolQ) / 70% (SciQ)** of the time.
- Errors are **largely inherited rather than independent** — so the rows needing correction are identifiable in principle.



Mechanism: recovery is ~linear in budget (volume-bound)

- Recovery of teacher-wrong rows is **~linear in budget** — 42% / 55% of the 0→100% gap recovered at 50%, tracking $y = x$ rather than a concave curve.
- A GT label's marginal value appears **roughly independent of which row it lands on**.
- Under volume-bound recovery, placement (*where*) and re-weighting (*how*) have **little leverage** — consistent with both nulls.

Mechanism — error recovery is ~LINEAR in budget (diffuse), not concave
→ targeting/combination can't help: only volume does



Power and controls

- **Power:** the paired per-(pair,seed) design gives **MDE = 0.0071**, below the 0.02 we'd treat as meaningful — so within-noise results are genuine nulls, not underpowered.
- **Pre-registration:** predictions git-anchored before the confirmatory seeds, and scored — including the refuted knee.
- **Exclusion:** gpt2-large GT is bimodal (an 8-seed run collapses below gpt2-medium 27% of the time) → excluded by a fixed rule, applied uniformly.
- **De-confounding:** a random-label control rules out training-set size — weak labels carry information (random < gt_only < mixing).
- **Cross-task:** the fraction curve, allocation null, and de-confound all replicate on **SciQ**.

Experiments across the three axes

Axis	Experiment	Outcome
How much	8-point fraction sweep, $0 \rightarrow 1$	back-loaded, saturates ~ 0.75
Where	error-targeting oracle + random control	null – within MDE
How	5 combination / loss methods	null – 1 above floor, still $< \text{xent}$
Loss	xent vs logconf	logconf inert $\rightarrow$ dropped
Tasks	BoolQ + SciQ	replicates on both
Mechanism	imitation-vs-correction probe	recovery $\sim$ linear in budget
Robustness	8-seed variance, seeds 3–4 reserve	exclusion by fixed rule

Coverage spans the three axes plus loss, task, and robustness checks; the negatives are concordant with the volume-bound account.

Takeaways, and the experiment that would change them

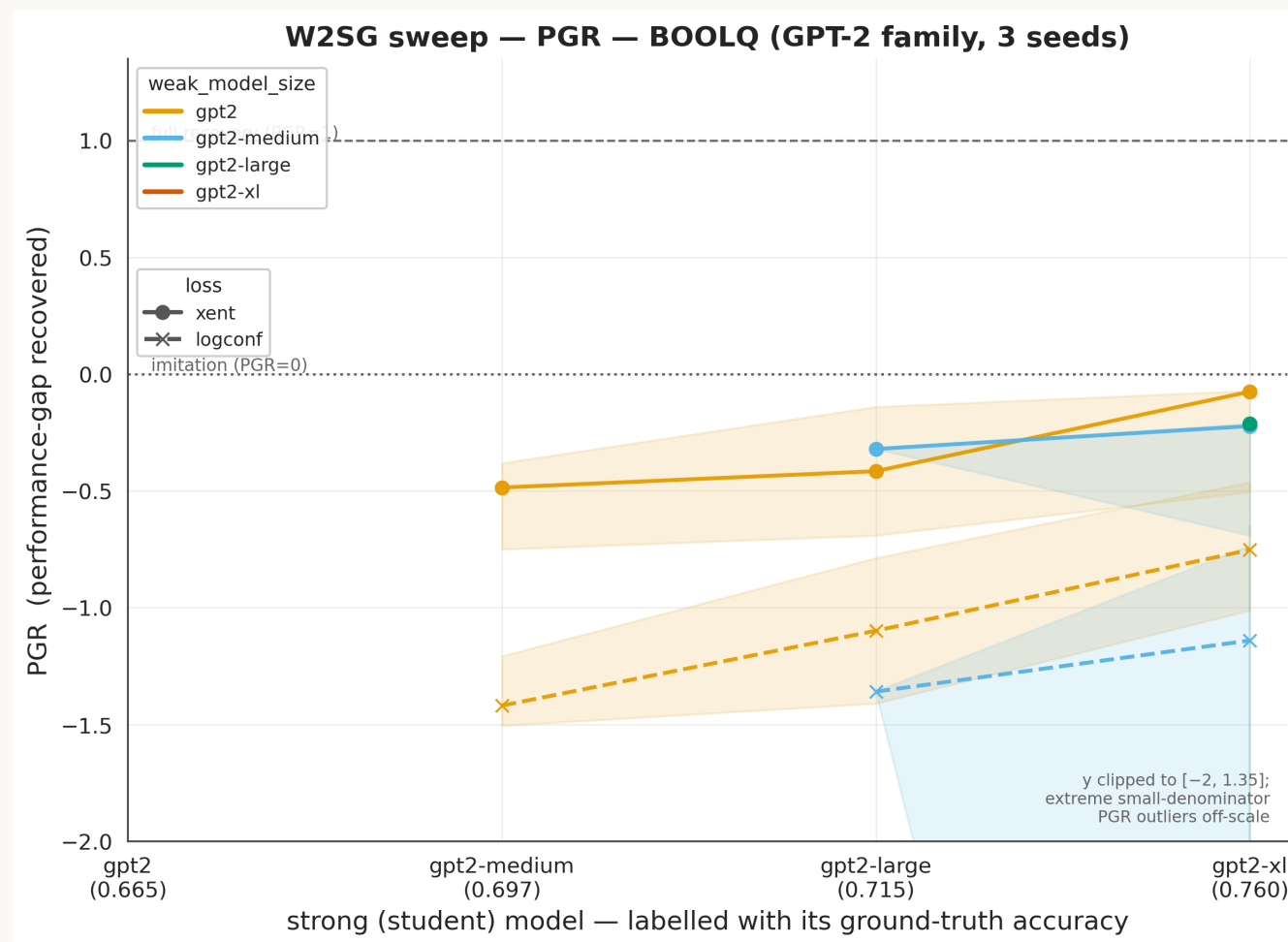
- Reproduced W2SG on the GPT-2 family and extended it to a supervision-budget setting.
- Decomposed the budget question into *how much* / *where* / *how* — a powered null on the last two, and a back-loaded, saturating curve on the first.
- One account — **inherited errors plus volume-bound recovery** — is consistent with all three.
- The main constraint here appears to be **scale**, not allocation or combination.
 - Direct test (out of scope under GPT-2-only): widen the student-teacher gap and re-run the sweep.
 - Falsifiable prediction: if recovery becomes **concave** at larger gaps, placement and combination should start to matter.

Appendix

frac=0 reproduction (acc + PGR) · transfer heatmap · per-method × pair detail · de-confound · cross-task

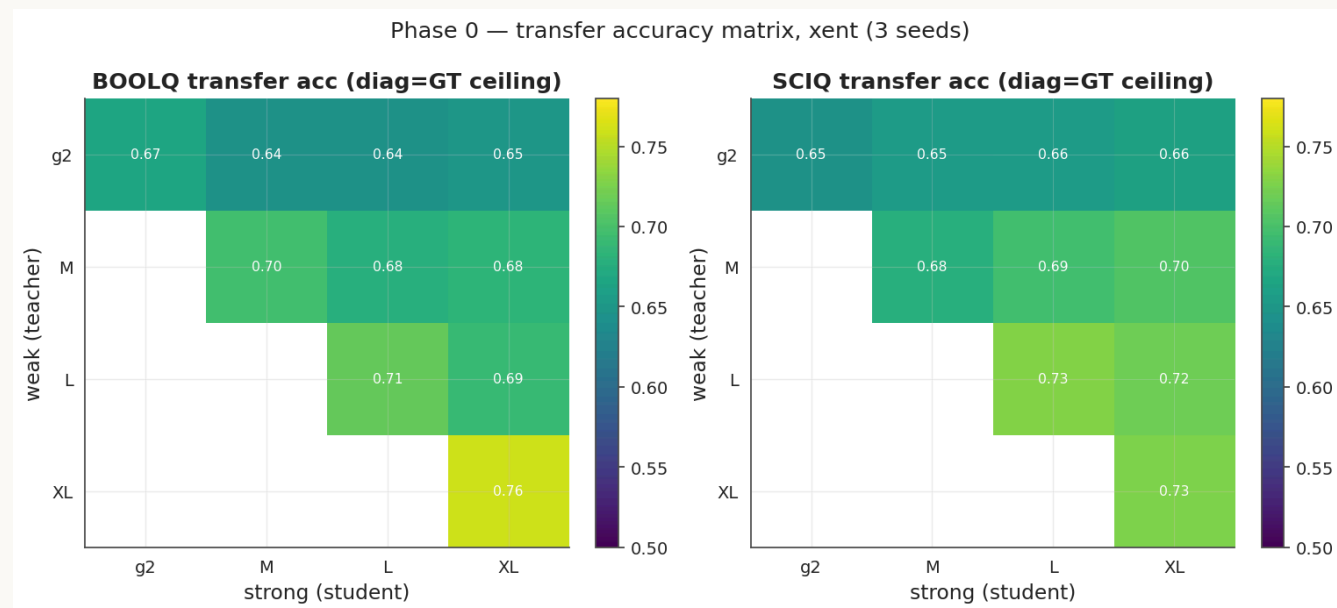
Appendix — frac=0 reproduction

- Standard W2SG sweep, GPT-2 family, BoolQ, **0% GT**, PGR axis.
- Median xent PGR **-0.27**; several pairs fall below the imitation line — BoolQ is low-signal at 0% GT.
- This is the baseline the 25%-GT sweep is measured against.



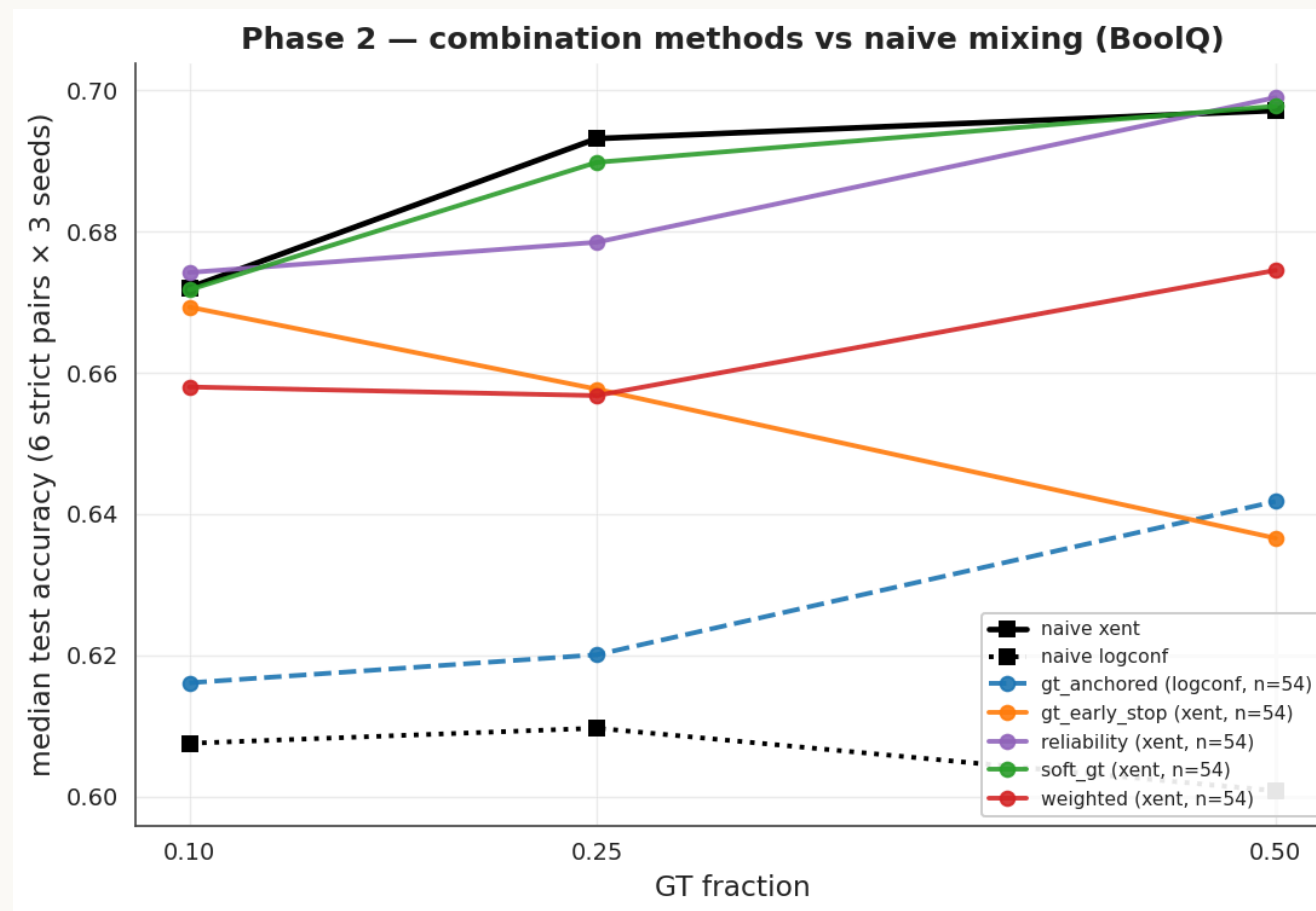
Appendix — transfer tracks the teacher more than student capacity

- With the teacher fixed, scaling the student adds $\sim +0.003$; with the student fixed, a better teacher adds $\sim +0.04$.
- Transfer tracks the teacher far more than student capacity — the static counterpart of the imitation result.



Appendix – combination portfolio, per-method detail

- All five methods track naive mixing across $\{0.10, 0.25, 0.50\}$.
- gt-anchored (logconf) is the only method above the floor; it recovers logconf but stays under xent.



Appendix — weak labels carry information (de-confound)

- Ordering: **random** < **gt_only** < **naive mixing** (15/15 BoolQ, 18/18 SciQ pairs).
- Replacing weak labels with noise reduces accuracy → the mixing gain reflects weak-label information, not just added rows.

